SLC-SOS 2026 Altitude Profiles (Version 1, created on 10/27/2026)

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Description: This document & accompanying spreadsheet detail the start & end times of all spirals & porpoises during the SLC-SOS 2026 campaign. This document contains altitude profiles, flight plans, instrument team notes when available, & smoke presence for all flights. There are 3 tabs in the spreadsheet: 1) Spirals based on altitude, 2) Spirals based on spatial information, & 3) Porpoises over the lake.

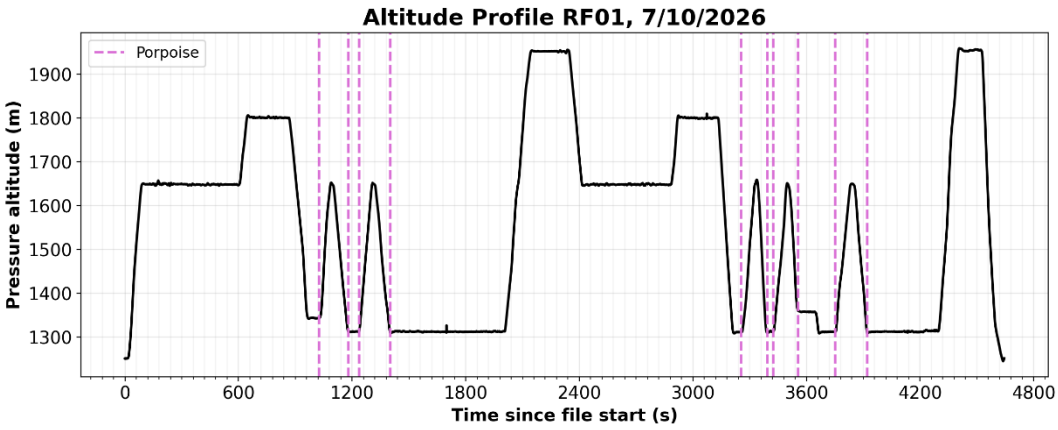
Spirals are identified visually by both sharp changes in altitude & through spatial visualization of when the aircraft enters the spiral area itself. For vertical profiling, the altitude-based profiles provide the most information, but for characterizing the site itself, the spatial-based profiles are more appropriate.

Porpoises were identified only by changes in altitude.

The time ranges for flights specify the start & end timestamps of the preliminary merged files (flight data c1), downloaded as available from August-September 2026, not the exact second of wheels up/down, though generally, this variation is less than 30 seconds.

Uncertainties: Start & end times were identified visually to the second, so an estimated uncertainty of ~3 seconds is expected.

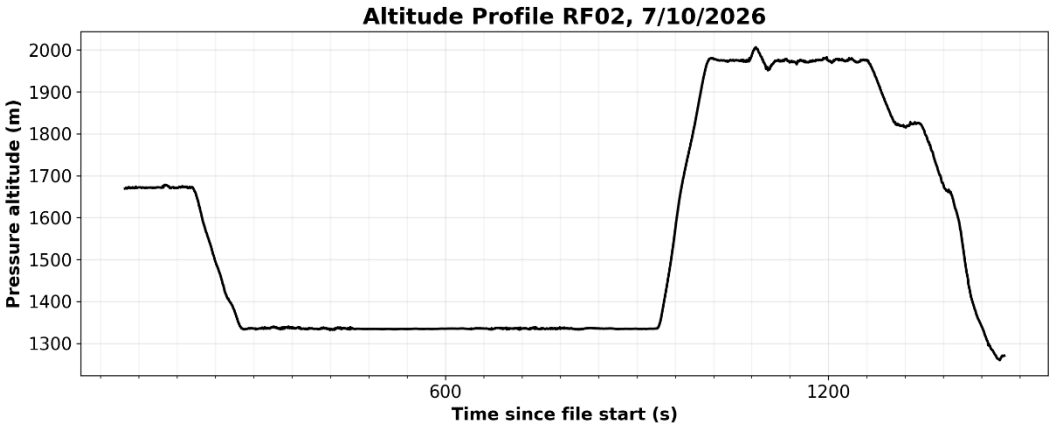
Summary for each Flight
Research flight 1 (RF01): 2026-07-10 15:15:12-16:32:37 UTC; 2 loops of morning downslope



Vertical Information

- 5 porpoises over the lake

Research flight 2 (RF02): 2026-07-10 19:41:30-20:06:05 UTC; 1 loop of afternoon lake breeze



Vertical Information

- None

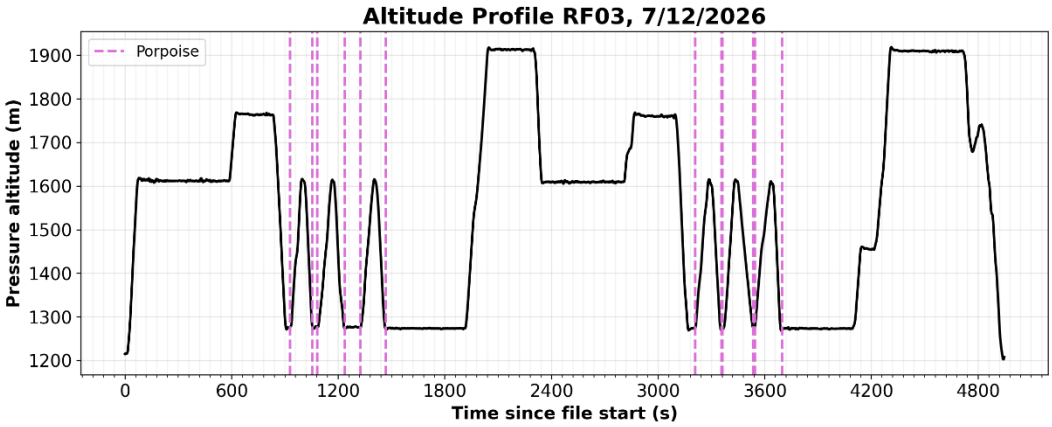
Instrumentation

- KA noseboom turned on after takeoff, so issues in wind & pressure variables at the start of flight

Notes

- Confusion with ATC led to a short flight due to instruments overheating
- Altitude incorrect for first 98 seconds of preliminary data

Research flight 3 (RF03): 2026-07-12 14:20:08-15:42:35 UTC; 2 loops of morning downslope



Vertical Information

- 6 porpoises over the lake

Instrumentation

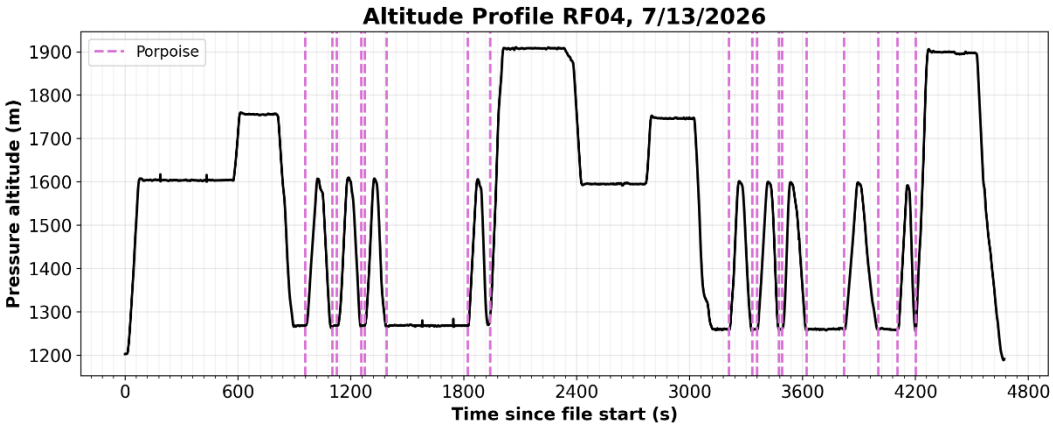
- PTR shut down at ~15:31 UTC
- HARP hardware error at 15:33 UTC & main power switch shut down at 15:37 UTC

- SP2-XR laser & pump shut down at 15:23:45 UTC

Notes

- Holding pattern started at 15:37 UTC
- HMS shows a dilute smoke plume

Research flight 4 (RF04): 2026-07-13 13:57:07-15:14:59 UTC; 2 loops of morning downslope



Vertical Information

- 9 porpoises over the lake

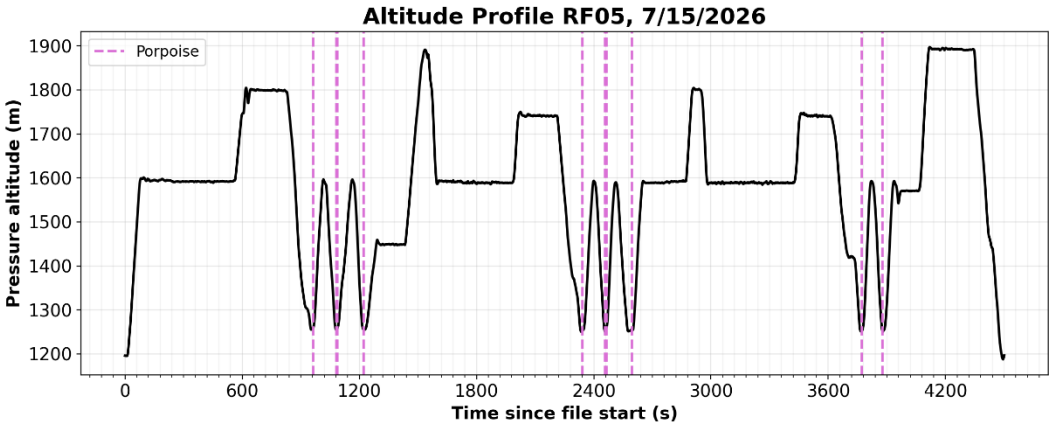
Instrumentation

- SP2-XR laser & pump shut down at 15:00 UTC
- Bypass pumps for SP2-XR & Aeris/Picarro turned on from 13:56:55-13:59:00 UTC during takeoff for vertical profile of light smoke

Notes

- Smoke observed

Research flight 5 (RF05): 2026-07-15 14:24:05-15:39:06 UTC; 3 loops of morning underpass



Vertical Information

- 5 porpoises over the lake

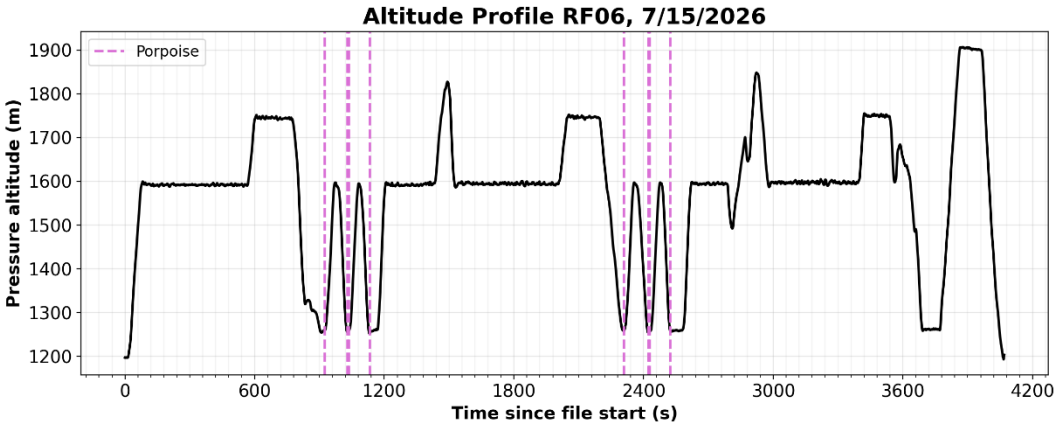
Instrumentation

- SP2-XR laser & pump shut down at 15:32:05 UTC

Notes

- Smoke observed

Research flight 6 (RF06): 2026-07-15 18:05:14-19:13:03 UTC; 2.5 loops of morning underpass



Vertical Information

- 4 porpoises over the lake

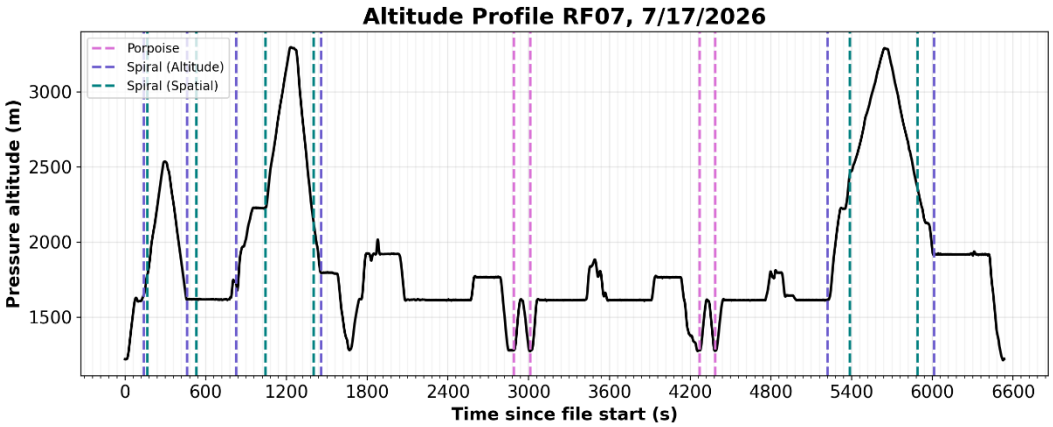
Instrumentation

- PTR shut down ~2 minutes before landing
- HARP reinitializing at 18:57 UTC
- SP2-XR shut down at 18:47 UTC

Notes

- Smoke observed
- TEMPO special operations

Research flight 7 (RF07): 2026-07-17 15:20:04-17:08:58 UTC; 4 loops of morning underpass



Vertical Information

- 3 spirals
    - Spiral 1 was over the University of Utah (Pandora Site)
    - Spirals 2 & 3 were over the Bingham Canyon Mine
- 2 porpoises over the lake

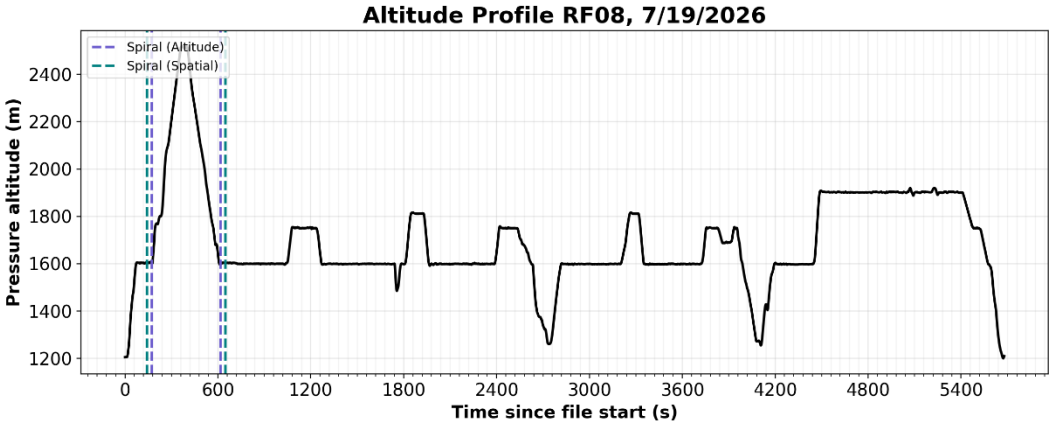
Instrumentation

- PTR shut down 3.5 loops in at ~16:48 UTC
- Only NO_2 data for CAPS
- HARP shut down at 17:02 UTC
- SP2-XR shut down 16:20 UTC

Notes

- TEMPO special operations

Research flight 8 (RF08): 2026-07-19 14:33:16-16:07:55 UTC; 3 loops of morning underpass



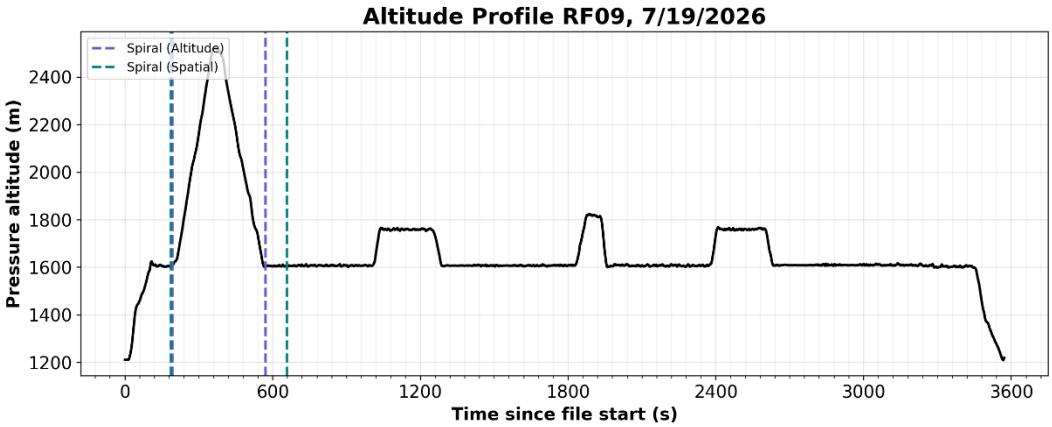
Vertical Information

- 1 spiral over the University of Utah

Instrumentation

- SP2-XR shut down at 15:30:50 UTC

Research flight 9 (RF09): 2026-07-19 18:21:40-19:21:11 UTC; 2 loops of morning underpass



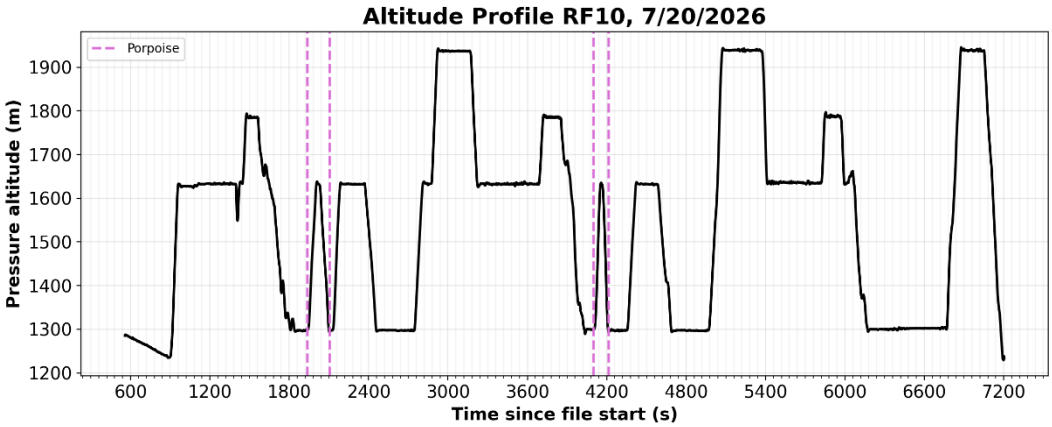
Vertical Information

- 1 spiral over the University of Utah

Instrumentation

- SP2-XR shut down at 19:08 UTC

Research flight 10 (RF10): 2026-07-20 15:16:19-17:16:23 UTC; ~2.75 loops of morning downslope



Vertical Information

- 2 porpoises over the lake

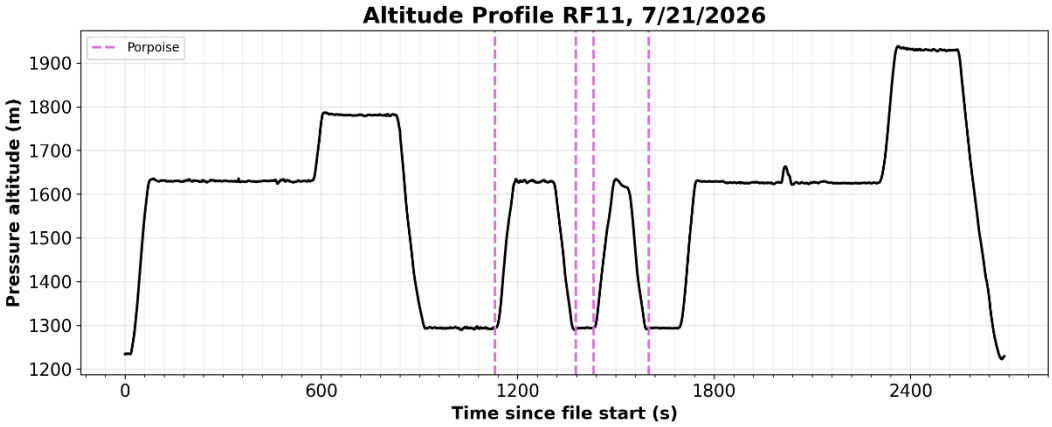
Instrumentation

- CAPS shut down at ~17:40 UTC
- SP2-XR shut down at 16:47:40 UTC

Notes

- Altitude incorrect for first 557 seconds of flight in preliminary data

Research flight 11 (RF11): 2026-07-21 14:39:20-15:24:06 UTC; 1 loop of morning downslope



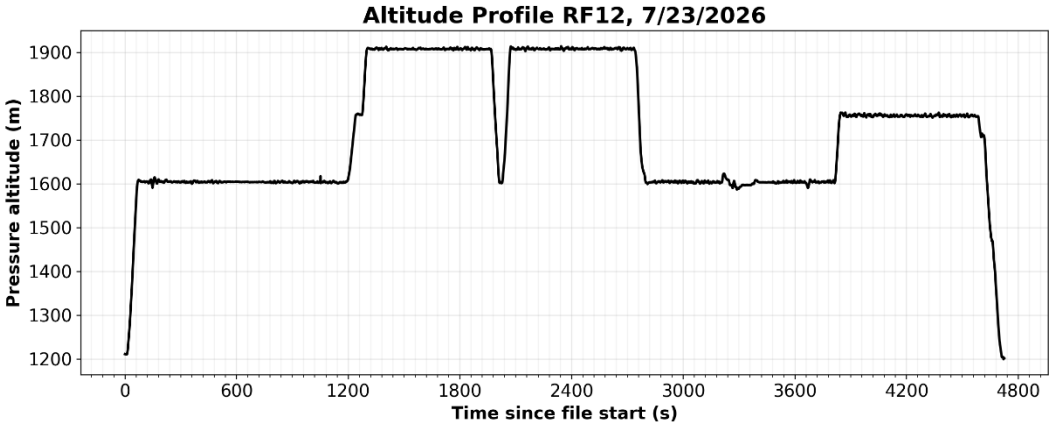
Vertical Information

- 2 porpoises over the lake

Instrumentation

- SP2-XR shut down at 14:54 UTC

Research flight 12 (RF12): 2026-07-23 16:24:35-17:43:20 UTC; 1.5 loops of intervalley



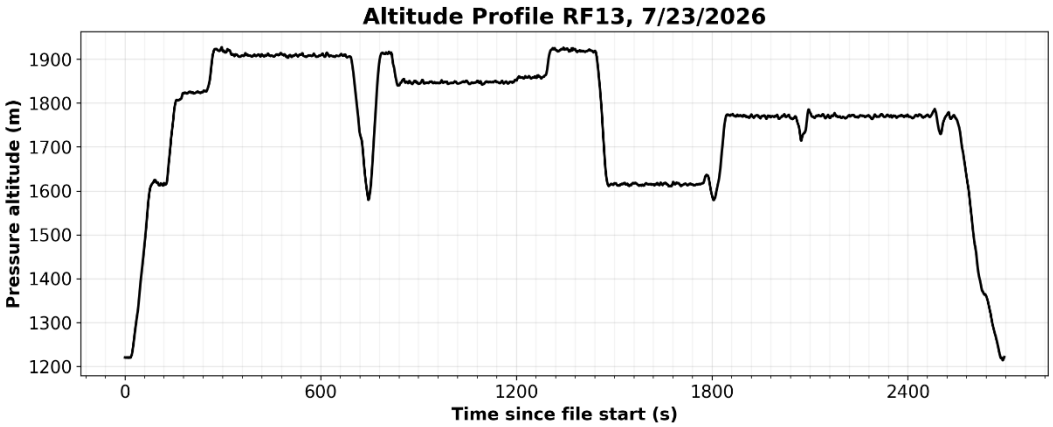
Vertical Information

- None

Notes

- Flew into smoke plume from a semi-truck fire at 17:38 UTC
- Holding pattern for landing started at 17:18 UTC

Research flight 13 (RF13): 2026-07-23 21:00:38-21:45:33 UTC; ~0.75 loops of intervalley



Vertical Information

- None

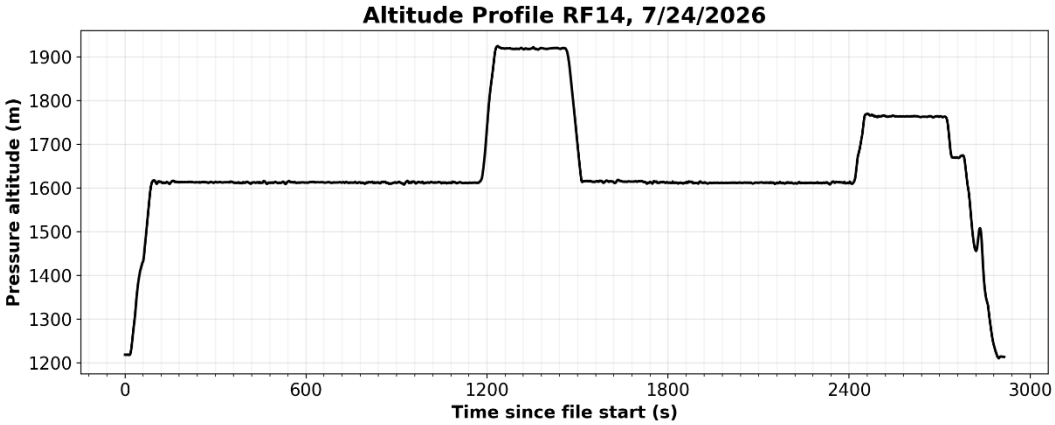
Instrumentation

- PTR shut down at ~19:28 UTC
- HARP restart at 20:48 UTC

Notes

- Holding pattern before landing

Research flight 14 (RF14): 2026-07-24 16:00:38-16:49:12 UTC; ~1.5 loops of intervalley



Vertical Information

- None

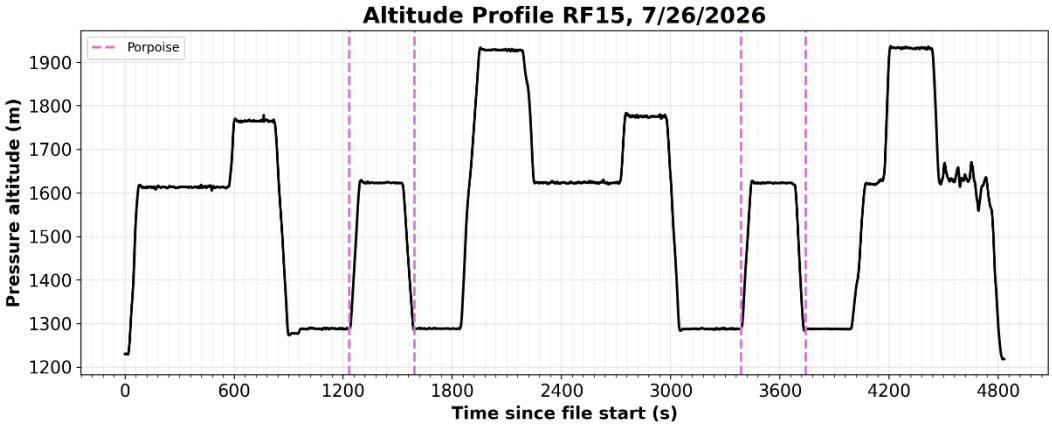
Instrumentation

- SP2-XR shut down at 16:35 UTC

Notes

- Northern loop cut short at Bountiful due to precipitation in the north

Research flight 15 (RF15): 2026-07-26 15:39:27-17:00:00 UTC; 2 loops of morning downslope



Vertical Information

- 2 porpoises over the lake

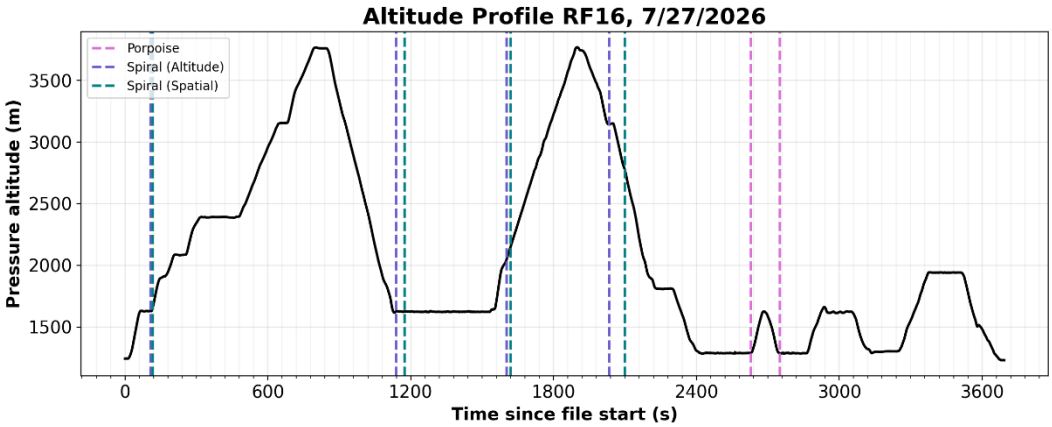
Instrumentation

- PTR signal loss at 16:07 UTC
- SP2-XR shut down at 16:34 UTC

Notes

- Holding pattern started at 16:51 UTC

Research flight 16 (RF16): 2026-07-27 15:46:10-16:47:43 UTC; 1 loop of morning downslope



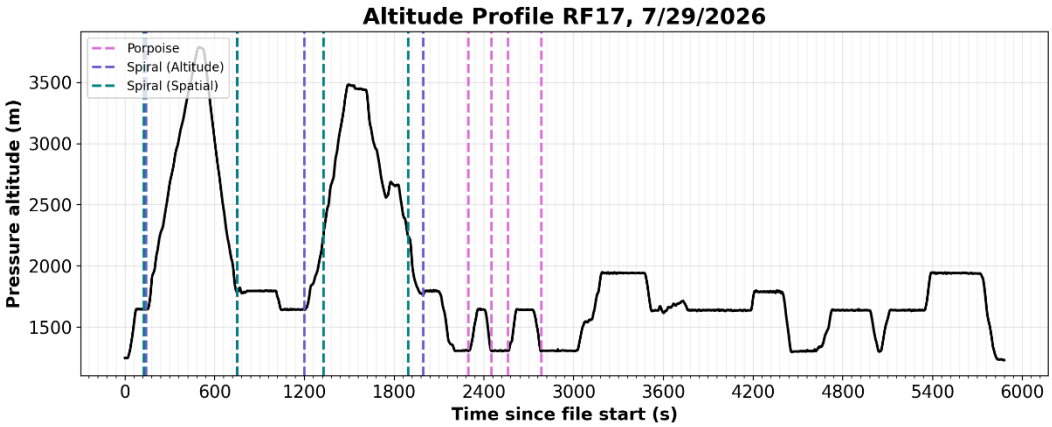
Vertical Information

- 2 spirals
    - Spiral 1 over the University of Utah
    - Spiral 2 over the Bingham Canyon Mine
- 1 porpoise over the lake

Instrumentation

- SP2-XR shut down at 16:37 UTC

<u>Research flight 17 (RF17):</u> 2026-07-29 16:43:45-18:21:44 UTC; 2 loops of morning downslope



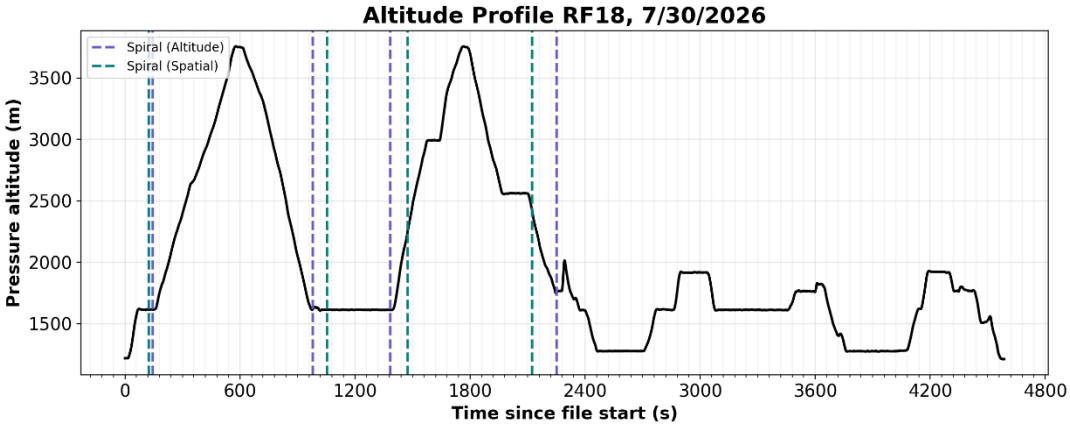
Vertical Information

- 2 spirals
  - Spiral 1 over the University of Utah
  - Spiral 2 over the Bingham Canyon Mine
- 2 porpoises over the lake

Notes

- Smoke observed

<u>Research flight 18 (RF18):</u> 2026-07-30 16:22:56-17:39:22 UTC; 2 loops of morning underpass



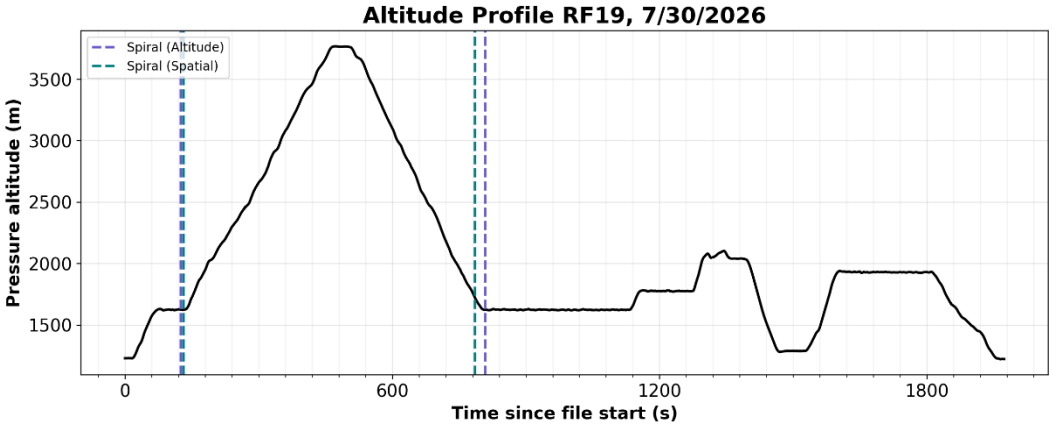
Vertical Information

- 2 spirals
  - Spiral 1 over the University of Utah
  - Spiral 2 over the Bingham Canyon Mine

Notes

- Smoke observed
- TEMPO special operations

Research flight 19 (RF19): 2026-07-30 20:15:06-20:47:59 UTC; 1 loop of morning underpass



Vertical Information

- 1 spiral over the University of Utah

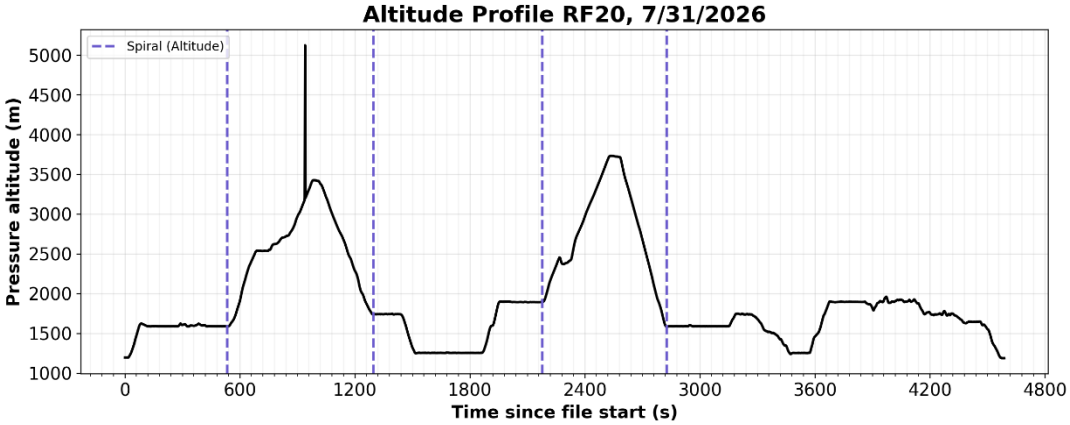
Instrumentation

- PTR signal drop about halfway through flight for ~60 seconds
- SP2-XR shut down during flight

Notes

- Smoke observed

Research flight 20 (RF20): 2026-07-31 16:11:55-17:28:23 UTC; 2 loops of morning underpass



Vertical Information

- 2 spirals
    - Spiral 1 over the Bingham Canyon Mine

    - Spiral 2 over the University of Utah

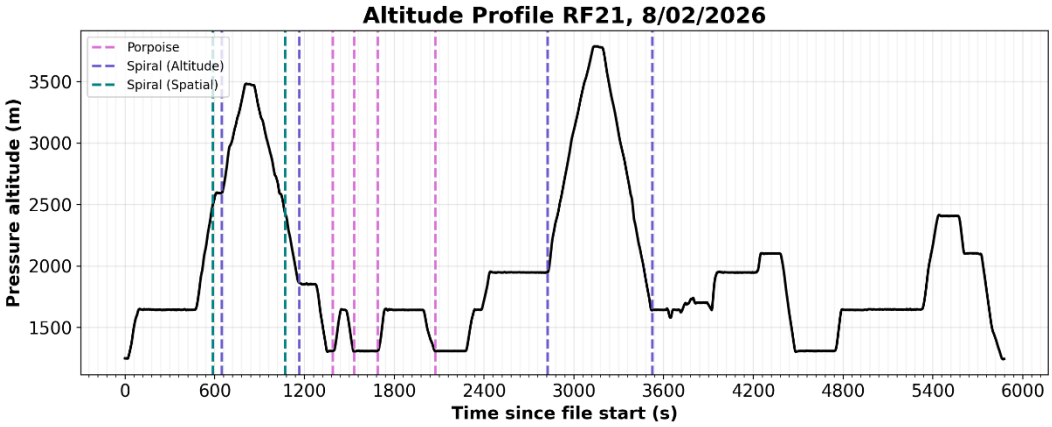
Instrumentation

- PTR signal drops at 16:55 & 17:13 UTC

Notes

- Light smoke observed (likely from Nevada fires)
- TEMPO special operations
- Pass through structure fire plume at ~16:14
- The Applanix was lost during the mine spiral in the preliminary data, so no spatial information available

Research flight 21 (RF21): 2026-08-02 16:11:17-17:49:14 UTC; 2 loops of morning downslope



Vertical Information

- 2 spirals
    - Spiral 1 over the Bingham Canyon Mine
    - Spiral 2 over the University of Utah
- 2 porpoises over the lake

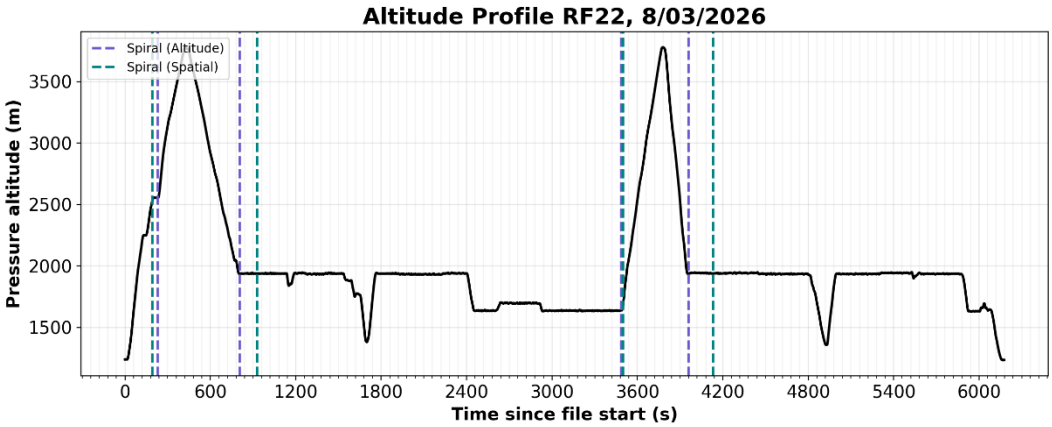
Instrumentation

- PTR signal drops at 16:59 UTC for 1 minute & 17:18 UTC for 2 minutes
- CAPS shut down at ~17:41 UTC

Notes

- Smoke observed
- Heading was lost from ~16:29-17:02 UTC, so no spatial information available for the University spiral

Research flight 22 (RF22): 2026-08-03 19:05:57-20:48:54 UTC; 1 loop of intervalley



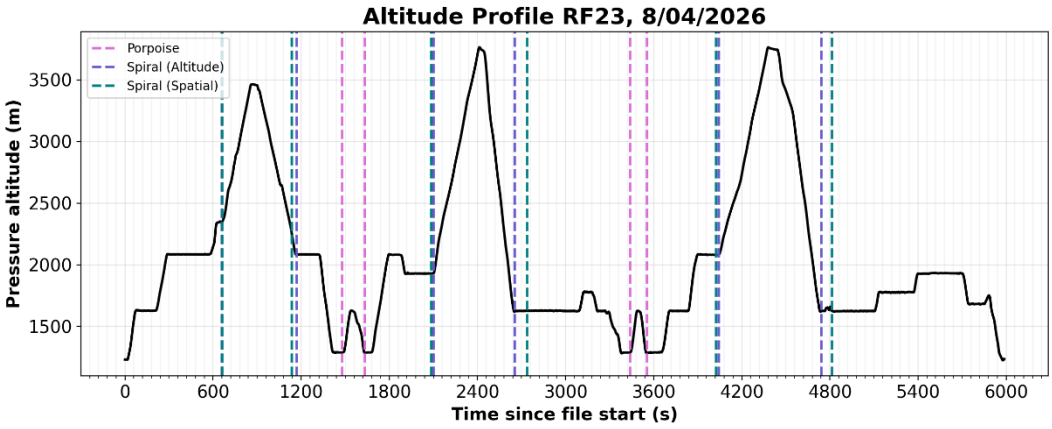
Vertical Information

- 2 spirals over the University of Utah

Notes

- Smoke observed

Research flight 23 (RF23): 2026-08-04 16:17:44-17:57:31 UTC; 2.5 loops of morning underpass



Vertical Information

- 3 spirals
    - Spiral 1 was over the Bingham Canyon Mine
    - Spirals 2 & 3 were over the University of Utah
- 2 porpoises over the lake

Instrumentation

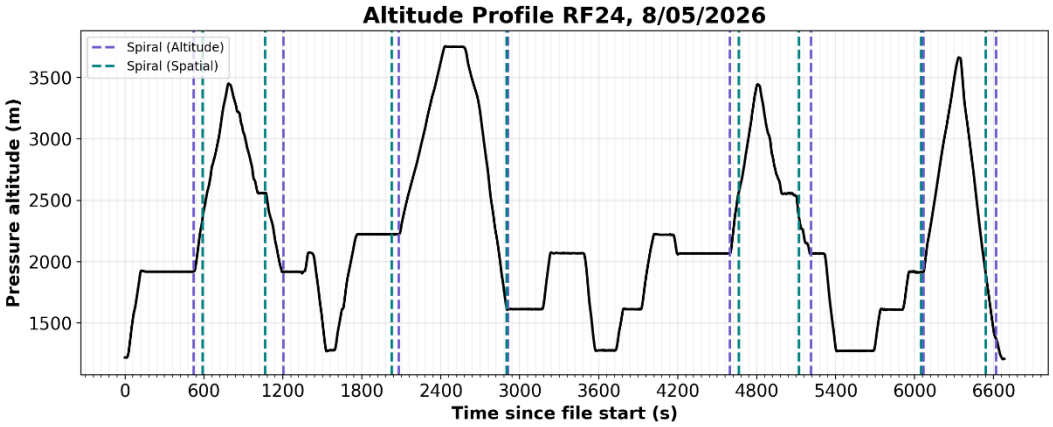
- PTR signal drops at 16:25 UTC for 45 seconds & 17:22 UTC for 3 minutes

Notes

- Smoke observed

- Holding pattern started at 17:47 UTC

<u>Research flight 24 (RF24):</u> 2026-08-05 16:09:19-18:00:40 UTC; 4 loops of morning underpass



Vertical Information

- 4 spirals
    - Spirals 1 & 3 were over the Bingham Canyon Mine
    - Spirals 2 & 4 were over the University of Utah

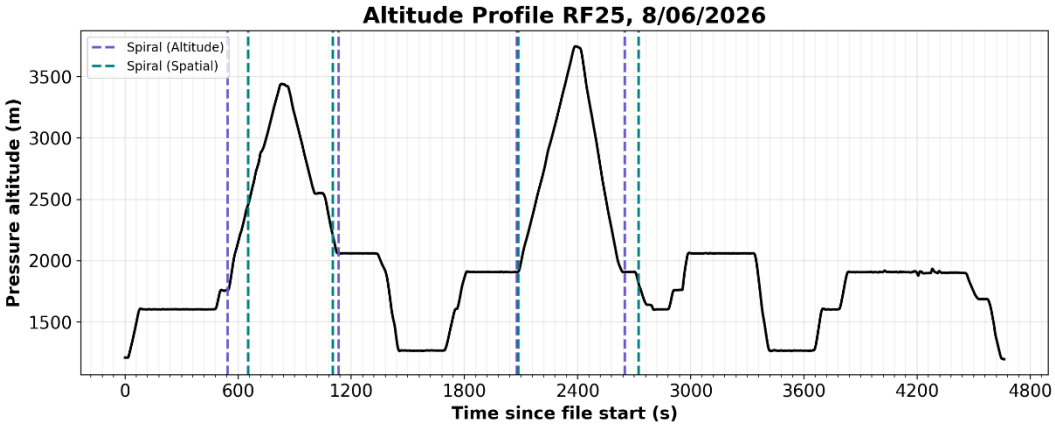
Instrumentation

- No PTR data

Notes

- Smoke observed

<u>Research flight 25 (RF25):</u> 2026-08-06 16:24:56-17:42:38 UTC; 2 loops of morning underpass



Vertical Information

- 2 spirals
    - Spiral 1 over the Bingham Canyon Mine
    - Spiral 2 over the University of Utah

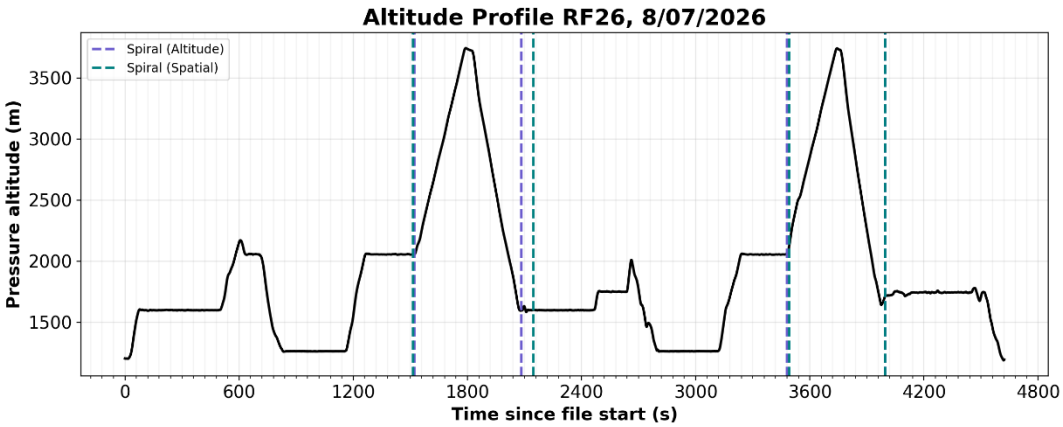
Instrumentation

- PTR signal drop at 16:28 UTC for 1 minute

Notes

- Smoke observed

Research flight 26 (RF26): 2026-08-07 16:20:32-17:37:34 UTC; 2 loops of morning underpass



Vertical Information

- 2 spirals over the University of Utah

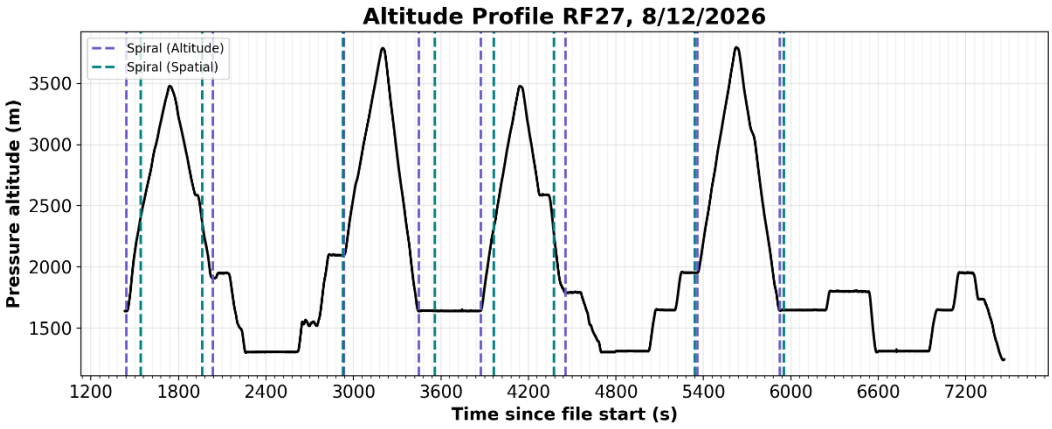
Instrumentation

- PTR signal drops at 16:33 UTC for 3 minutes, 16:53 UTC for 5 minutes, 17:06 UTC for 5 minutes, 17:25 UTC for 6 minutes, & during landing

Notes

- Smoke observed
- TEMPO special operations

Research flight 27 (RF27): 2026-08-12 15:58:16-18:02:39 UTC; 3 loops of morning underpass



Vertical Information

- 4 spirals
    - Spirals 1 & 3 were over the Bingham Canyon Mine
    - Spirals 2 & 4 were over the University of Utah

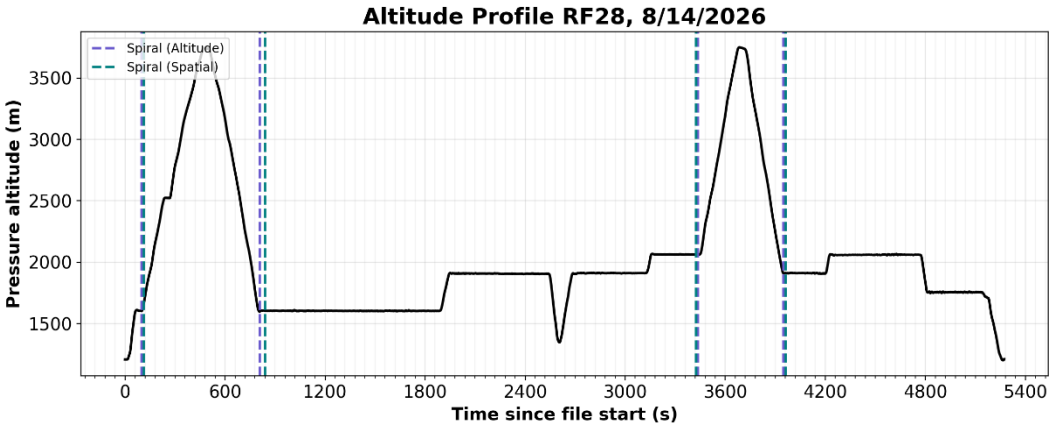
Instrumentation

- PTR signal drop at 16:35 UTC for 1 minute
- HARP out for ~half of the flight
- No temperature or wind data for the first 6 minutes of flight

Notes

- Some smoke observed
- Altitude incorrect for first 1432 seconds of file in preliminary data
    - Based on instrument team notes, this likely only impacts the first ~8 minutes of the flight itself

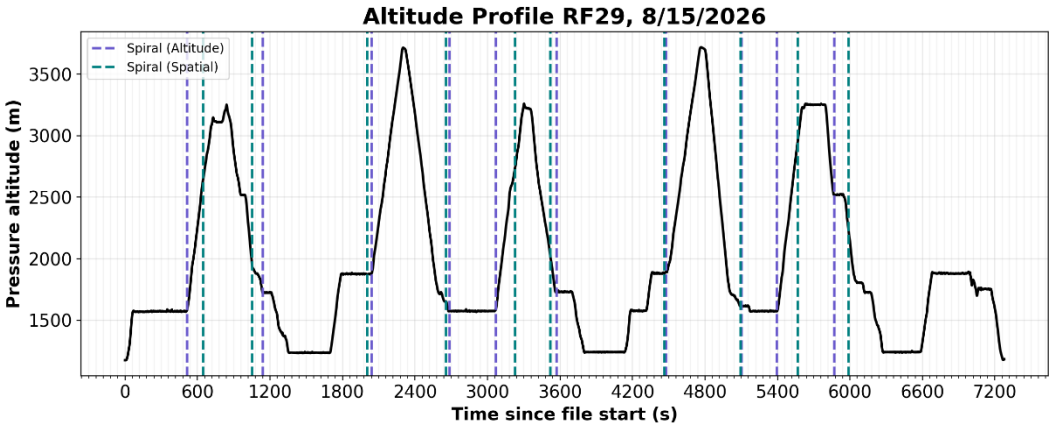
<u>Research flight 28 (RF28):</u> 2026-08-14 15:31:52-16:59:45 UTC; 1.5 loops of intervalley



Vertical Information

- 2 spirals over the University of Utah

Research flight 29 (RF29): 2026-08-15 19:07:56-21:09:13 UTC; 3 loops of morning underpass



Vertical Information

- 5 spirals
    - Spirals 1, 3, & 5 were over the Bingham Canyon Mine
    - Spirals 2 & 4 were over the University of Utah

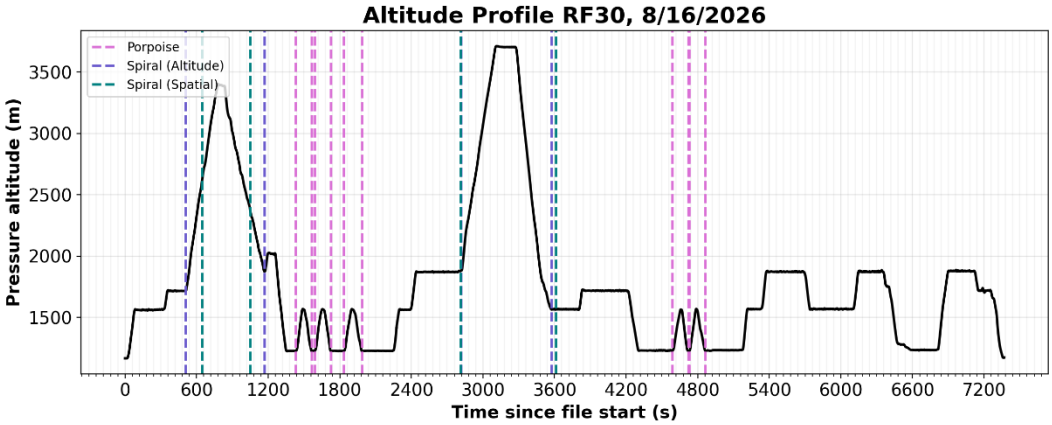
Instrumentation

- No PTR data

Notes

- TEMPO special operations
- Holding pattern started at 20:43 UTC

Research flight 30 (RF30): 2026-08-16 17:36:26-19:39:15 UTC; 2.5 loops of morning underpass



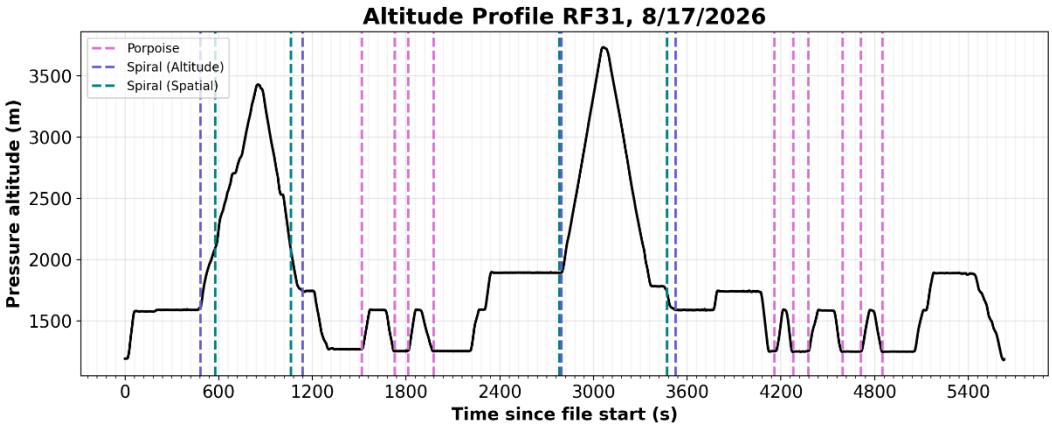
Vertical Information

- 2 spirals
    - Spiral 1 was over the Bingham Canyon Mine
    - Spiral 2 was over the University of Utah
- 5 porpoises over the lake

Instrumentation

- Picarro issues ~18:49 UTC

<u>Research flight 31 (RF31):</u> 2026-08-17 15:30:33-17:04:23 UTC; 2 loops of morning downslope



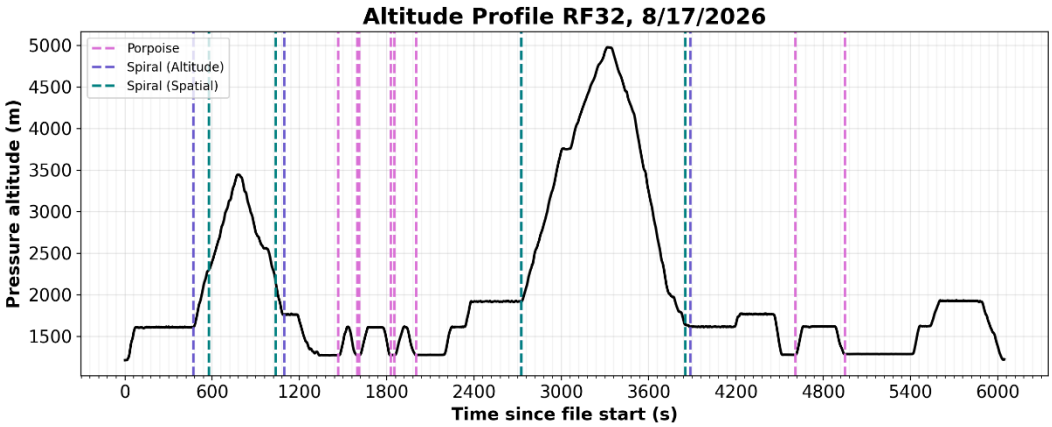
Vertical Information

- 2 spirals
  - Spiral 1 over the Bingham Canyon Mine
  - Spiral 2 over the University of Utah
- 5 porpoises over the lake

Instrumentation

- PTR signal drops at 16:20 UTC for 2 minutes, 16:30 UTC for 2 minutes, & 16:42 UTC for 10 minutes
- SP2-XR shut down at 16:44 UTC

<u>Research flight 32 (RF32):</u> 2026-08-17 20:37:18-22:18:03 UTC; 2 loops of morning downslope



Vertical Information

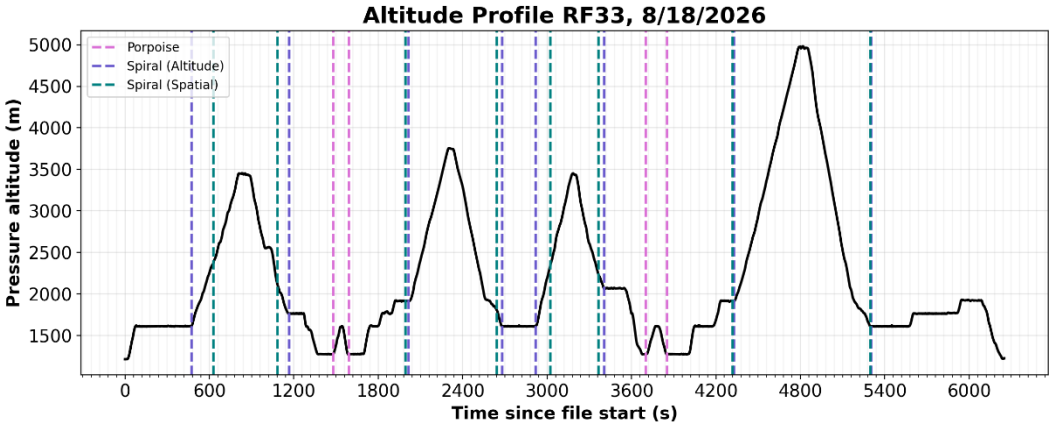
- 2 spirals
  - Spiral 1 over the Bingham Canyon Mine

    - Spiral 2 over the University of Utah
- 4 porpoises over the lake

Instrumentation

- PTR signal drop from 22:05 UTC to end of flight
- CAPS pressure controller issue at ~17,500 ft, but data corrected for NO_2
    - Still an issue with NO_x in the preliminary data
- CAPS shut down a few minutes before landing
- SP2-XR shut down at 21:44 UTC

Research flight 33 (RF33): 2026-08-18 18:38:04-20:22:13 UTC; 2.5 loops of morning underpass



Vertical Information

- 4 spirals
    - Spirals 1 & 3 were over the Bingham Canyon Mine
    - Spirals 2 & 4 were over the University of Utah
- 2 porpoises over the lake

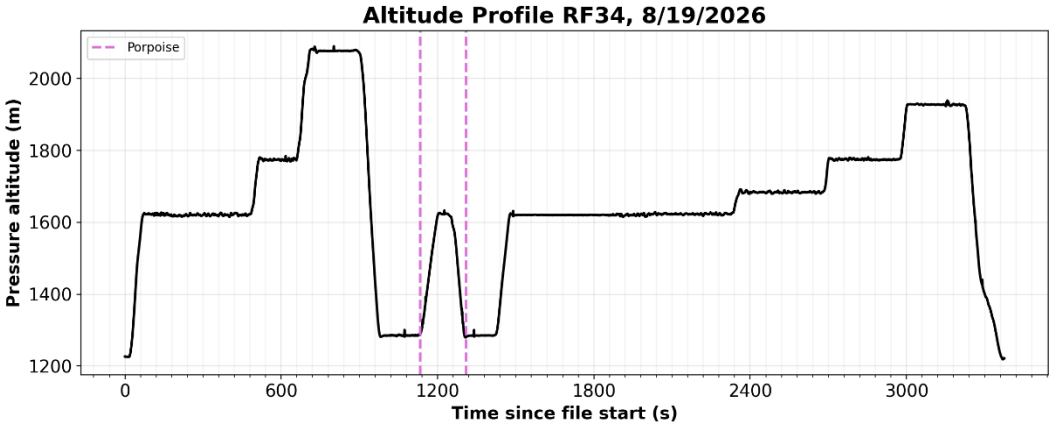
Instrumentation

- PTR signal drops from 19:12-19:21 UTC & 19:54 UTC to end of flight
- SP2-XR shut down at 20:13 UTC

Notes

- TEMPO special operations

Research flight 34 (RF34): 2026-08-19 20:08:31-21:04:46 UTC; 1.5 loops of morning downslope



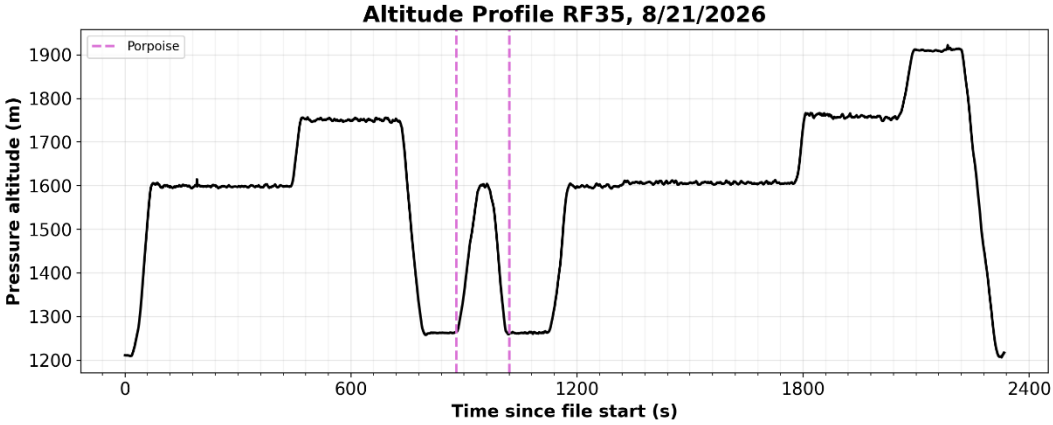
Vertical Information

- 1 porpoise over the lake

Instrumentation

- PTR signal drop from 21:02 UTC to end of flight
- SP2-XR shut down at 20:46 UTC

Research flight 35 (RF35): 2026-08-21 20:15:25-20:54:19 UTC; 1 loop of morning underpass



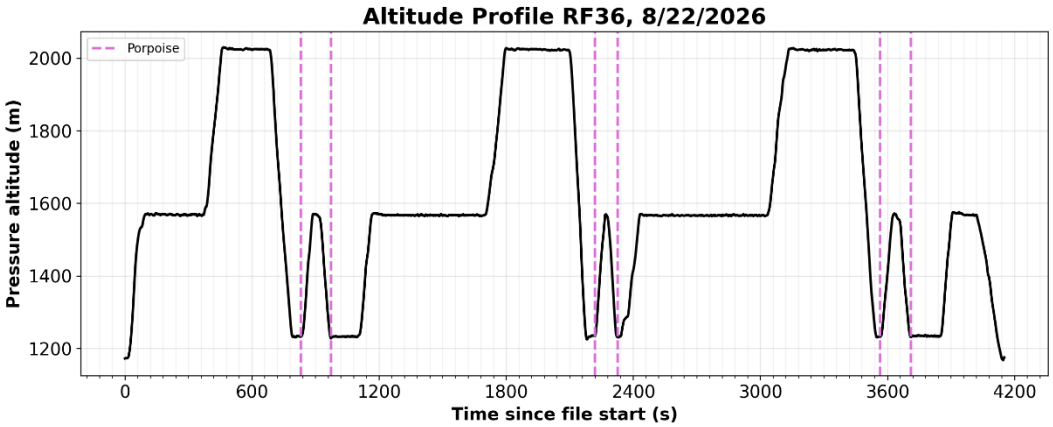
Vertical Information

- 1 porpoise over the lake

Instrumentation

- PTR shut down at 20:41 UTC
- SP2-XR shut down at 20:27 UTC

Research flight 36 (RF36): 2026-08-22 16:21:57-17:31:08 UTC; 3 loops of morning underpass



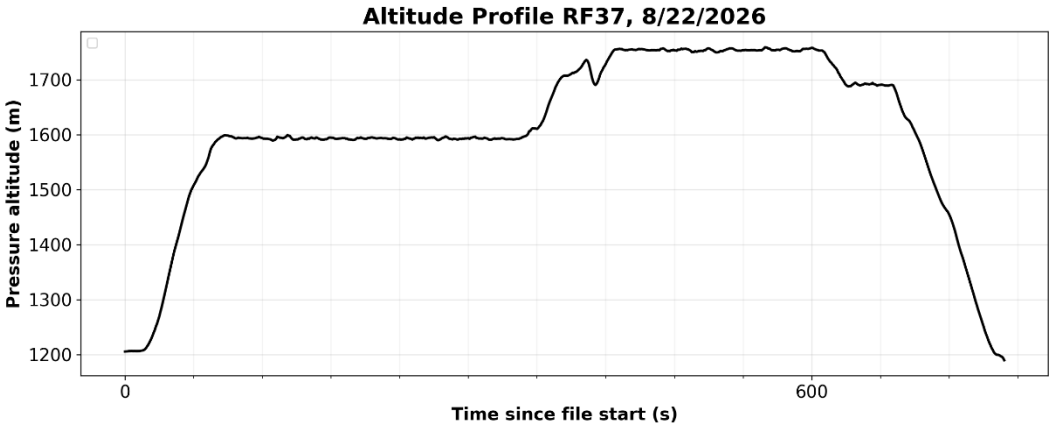
Vertical Information

- 3 porpoises over the lake

Instrumentation

- SP2-XR shut down at 17:18 UTC

Research flight 37 (RF37): 2026-08-22 20:59:31-21:12:19 UTC; <1 loop of morning underpass



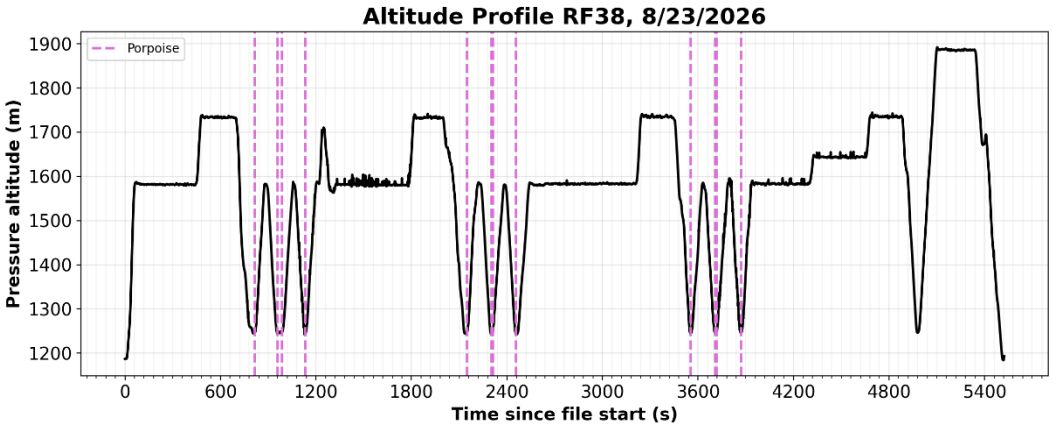
Vertical Information

- None

Notes

- Flight scrubbed just after takeoff due to an unavoidable thunderstorm in the southwest

Research flight 38 (RF38): 2026-08-23 16:29:44-18:01:48 UTC; 3.5 loops of morning underpass



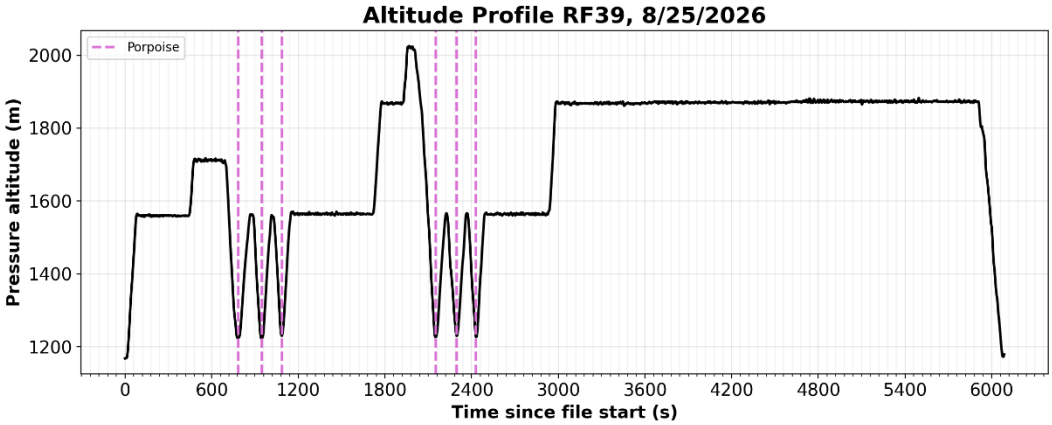
Vertical Information

- 6 porpoises over the lake

Instrumentation

- SP2-XR shut down at 17:49 UTC

Research flight 39 (RF39): 2026-08-25 17:50:29-19:31:56 UTC; 2 loops of morning underpass & 3 urban boxes over southern half



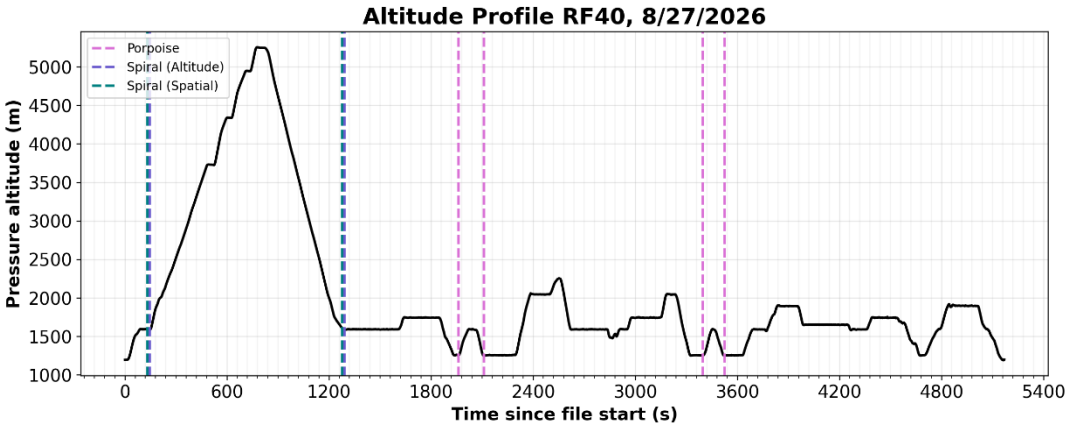
Vertical Information

- 4 porpoises over the lake

Instrumentation

- PTR shut down after 2 urban box loops
- SP2-XR shut down at 19:07:20 UTC

Research flight 40 (RF40): 2026-08-27 17:04:03-18:30:12 UTC; 2.5 loops of morning underpass



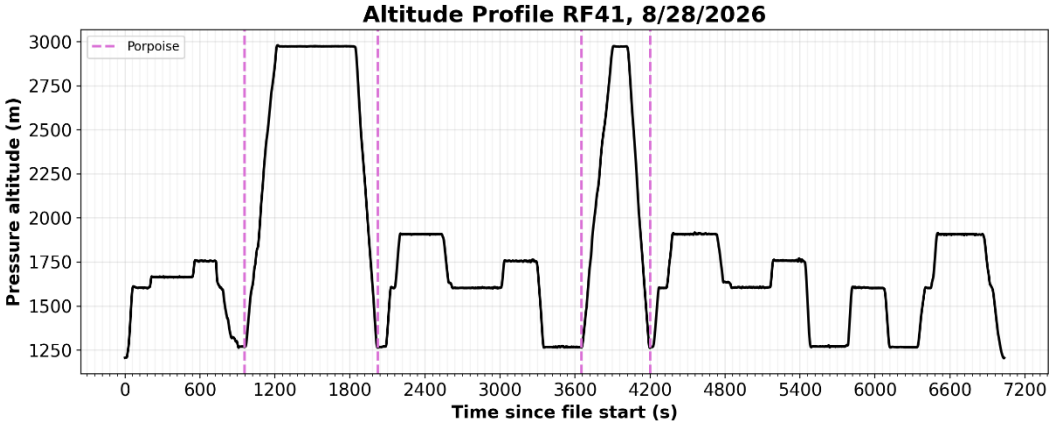
Vertical Information

- 1 spiral over the University of Utah
- 2 porpoises over the lake

Instrumentation

- PTR shut down at 18:20 UTC
- CAPS data issues above 14,000 ft
- SP2-XR shut down at 18:08:40 UTC

Research flight 41 (RF41): 2026-08-28 16:59:21-18:56:35 UTC; 3 loops of morning downslope



Vertical Information

- 2 large porpoises/traffic cone soundings over the lake

Instrumentation

- PTR shut down at 18:55 UTC
- SP2-XR shut down at 18:28 UTC